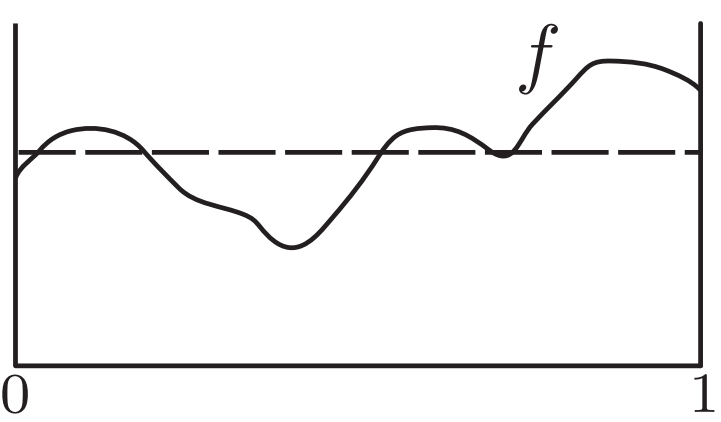
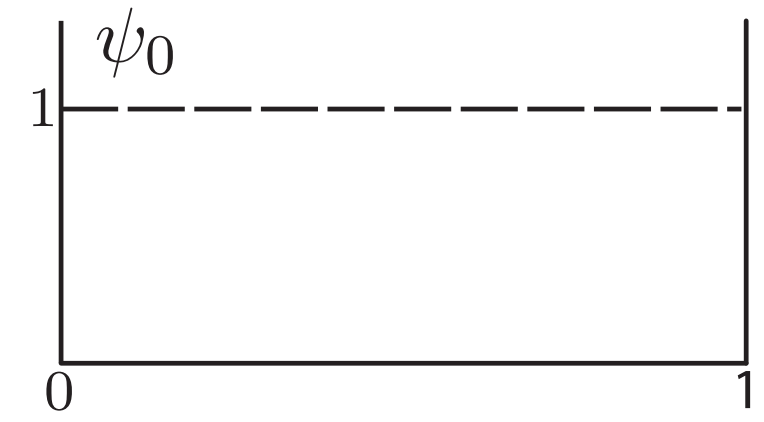
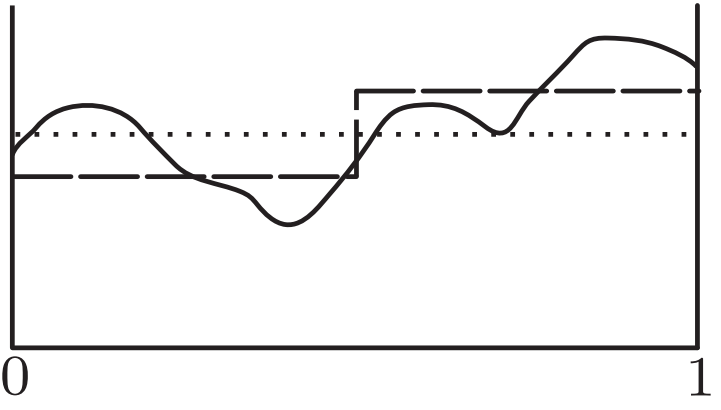
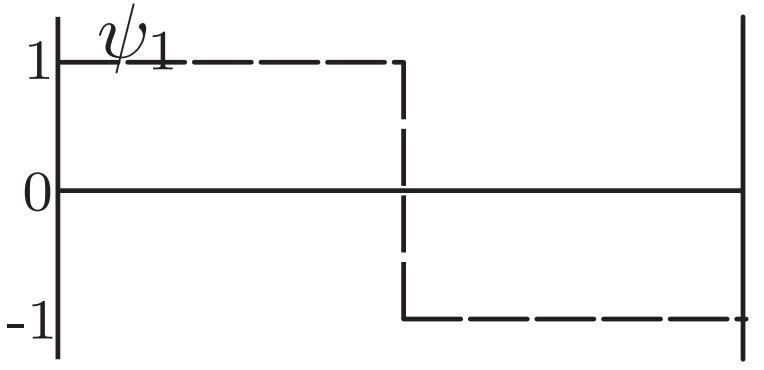


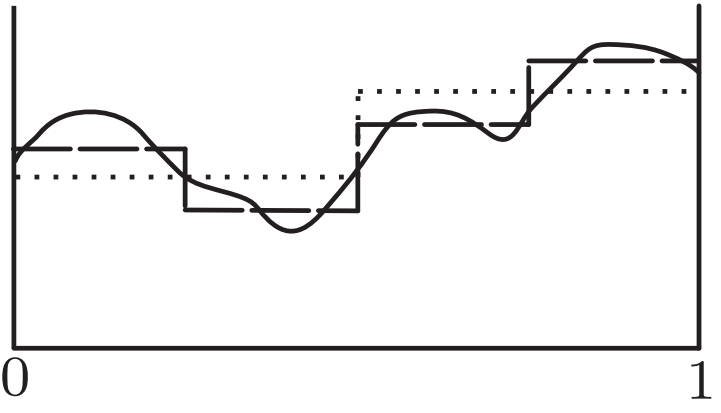
$$f = \langle \psi_0, f \rangle \psi_0$$





$$+\langle \psi_1, f \rangle \psi_1$$





$$+\langle \psi_2, f \rangle \psi_2 + \langle \psi_3, f \rangle \psi_3 + \dots$$